

MESHIFY

NETWORK SCIENCE PROJECT

Guneet Pal Singh(2022190)
Madhav Kansil(2022270)
Harsh Kumar(2022198)



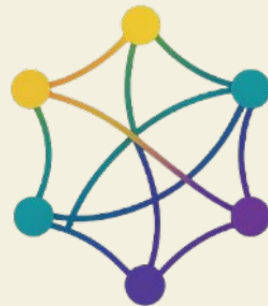
1. **Introduction**
2. **Datasets Used and Work Done**
3. **Drawing a Conclusion**

1. **Introduction**
2. **Datasets Used and Work Done**
3. **Drawing a Conclusion**

MESHIFY

Mapping the Dynamic relationship between artists and genres

This framework models the fluid connections between artists and genres by analyzing social behavior data, such as playlist co-occurrences, to assign multi-genre affinity scores instead of rigid labels. It visualizes genres as interconnected nodes in an evolving network, reflecting historical lineage, cross-genre influences, and community-driven associations. By quantifying artists' "bridge" roles between genres and mapping correlations like Rap-R&B fusion or Country-Reggae exclusivity, it captures the nuanced, dynamic nature of musical identity beyond traditional taxonomies.



1. Introduction
2. **Datasets Used and Work Done**
3. Drawing a Conclusion

Datasets Used

We used 2 Spotify Datasets in total.

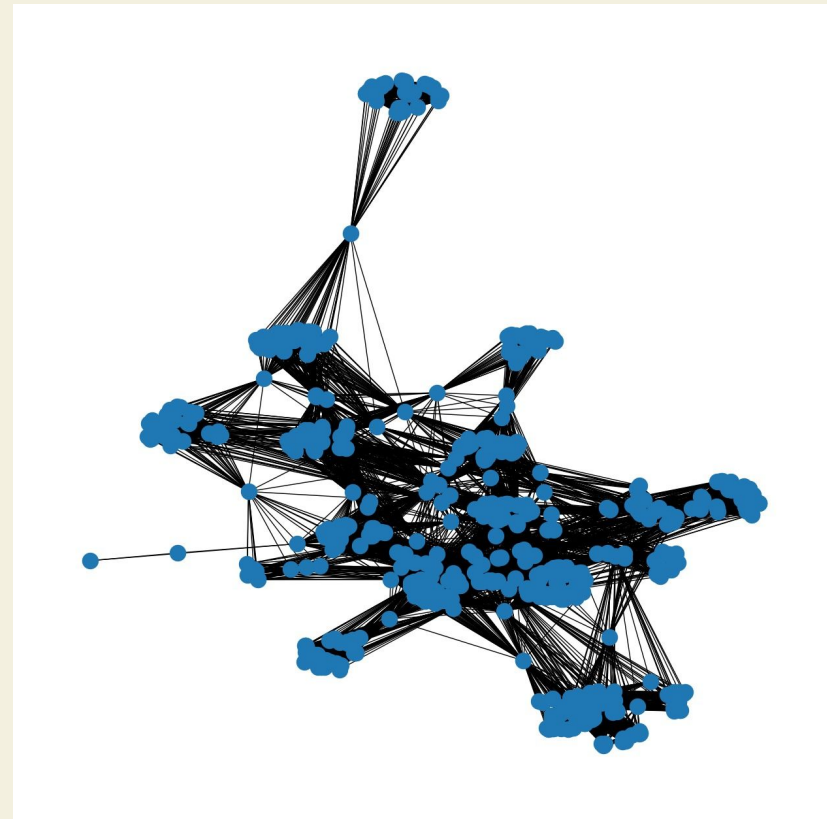
The link of each database is given below

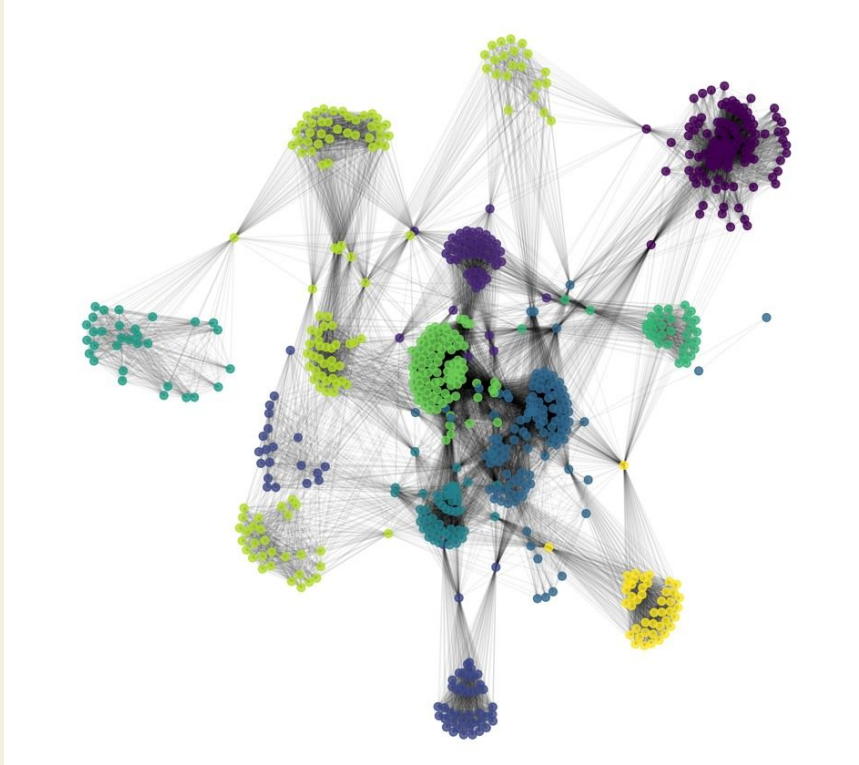
- 1) MPD spotify dataset – <https://raw.githubusercontent.com/vaslnk/Spotify-Song-Recommendation-ML/refs/heads/master/data/mpd.slice.0-999.json>
- 2) Spotify Personal Database – Used exportify to extract the database on our own spotify usage. <https://exportify.net/>



Dataset 1 EDA

This graph shows the connections between playlists in our sampled database. Each blue dot is a playlist, and the black lines represent shared songs or similarities. Dense clusters indicate groups of playlists with many tracks in common, while isolated dots have fewer connections. The dataset was reduced from about **one million data points to 4,000 playlists** for easier analysis.

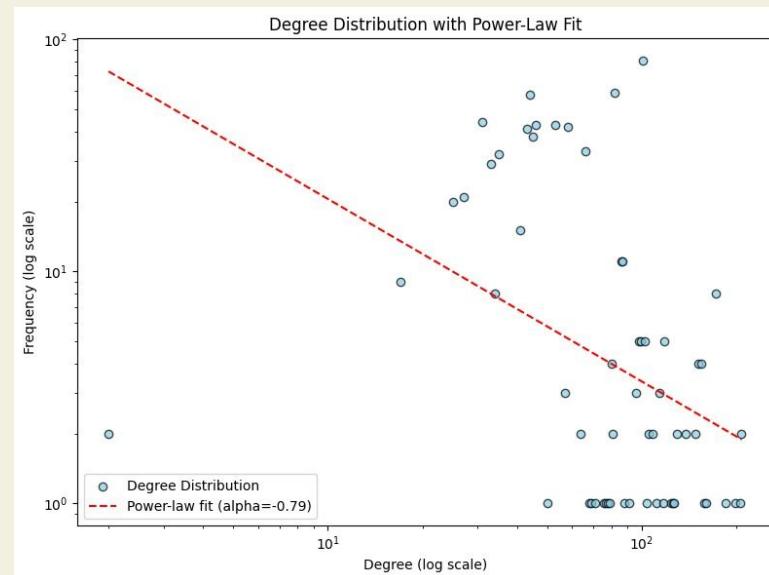
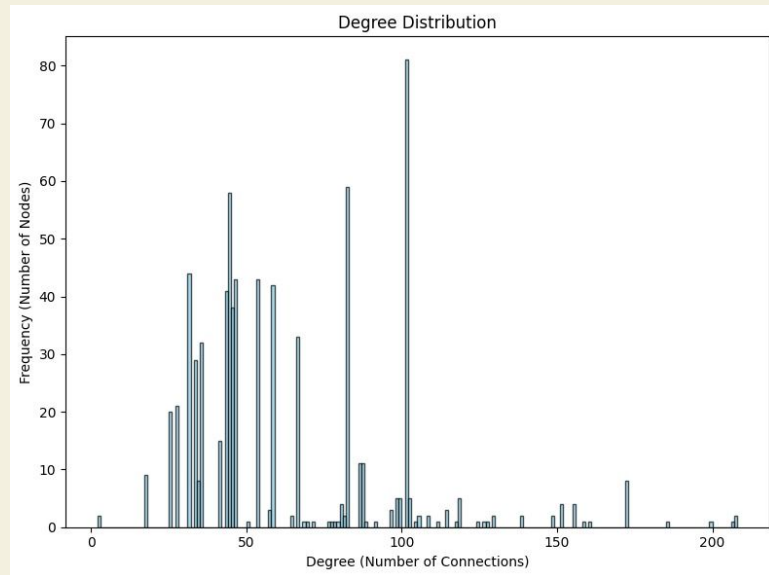




Communities

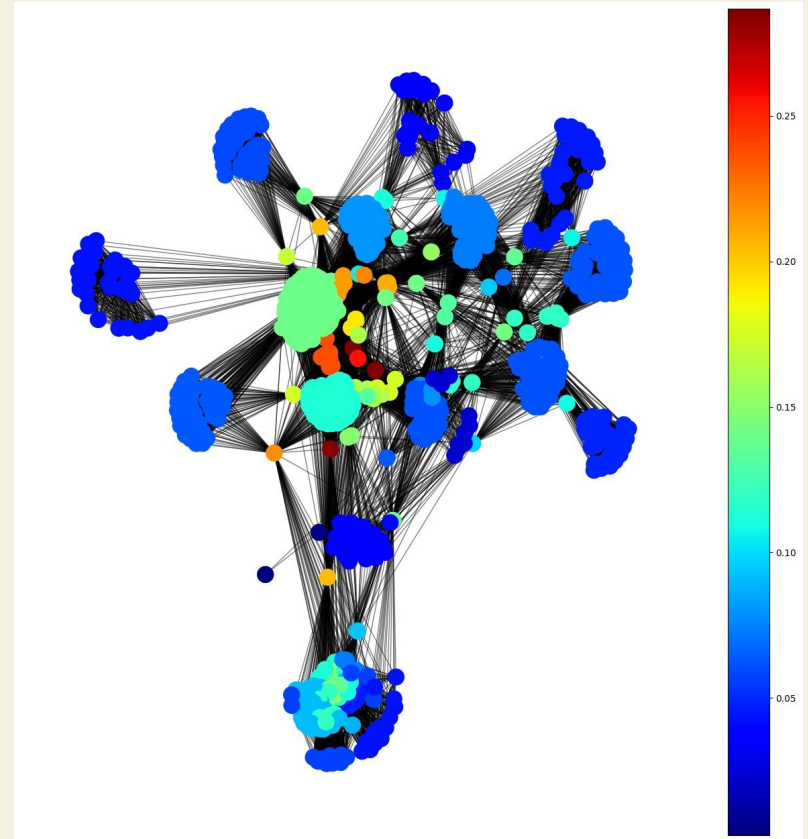
This graph visualizes the community structure within Dataset 1. It shows **10 distinct communities**, each grouped by color. The network's **diameter is 5**, meaning the longest shortest path between any two nodes is 5 steps, and **the radius is 3**, indicating a central node is at most 3 steps from all others. The clusters highlight strong internal connections within each community and fewer links between communities

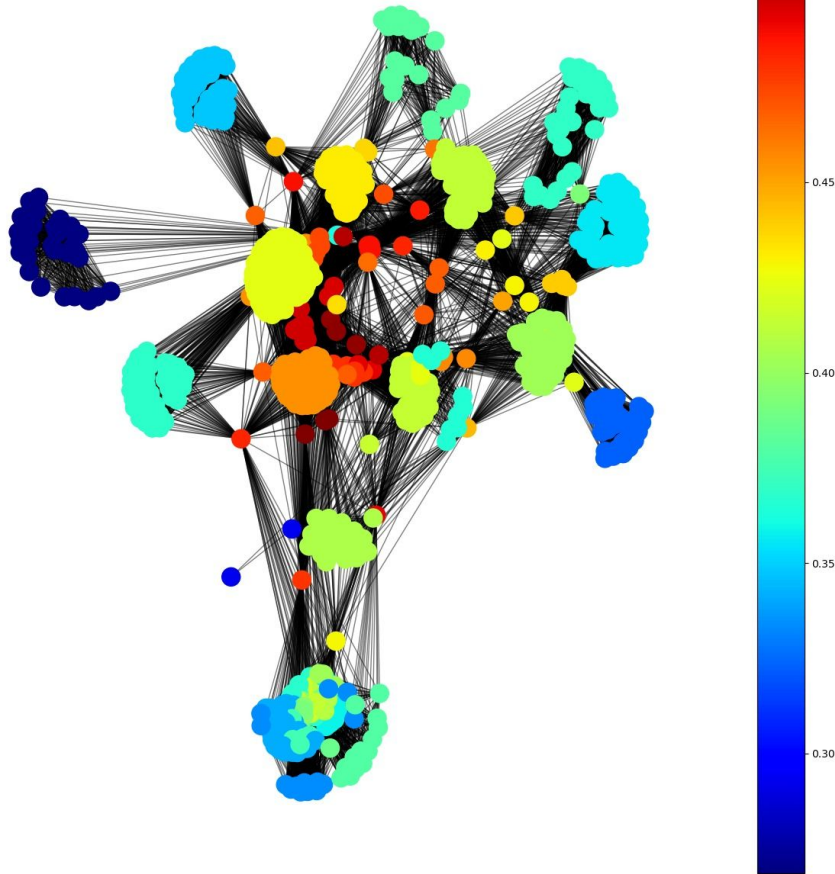
Degree Distributions



Degree Centrality

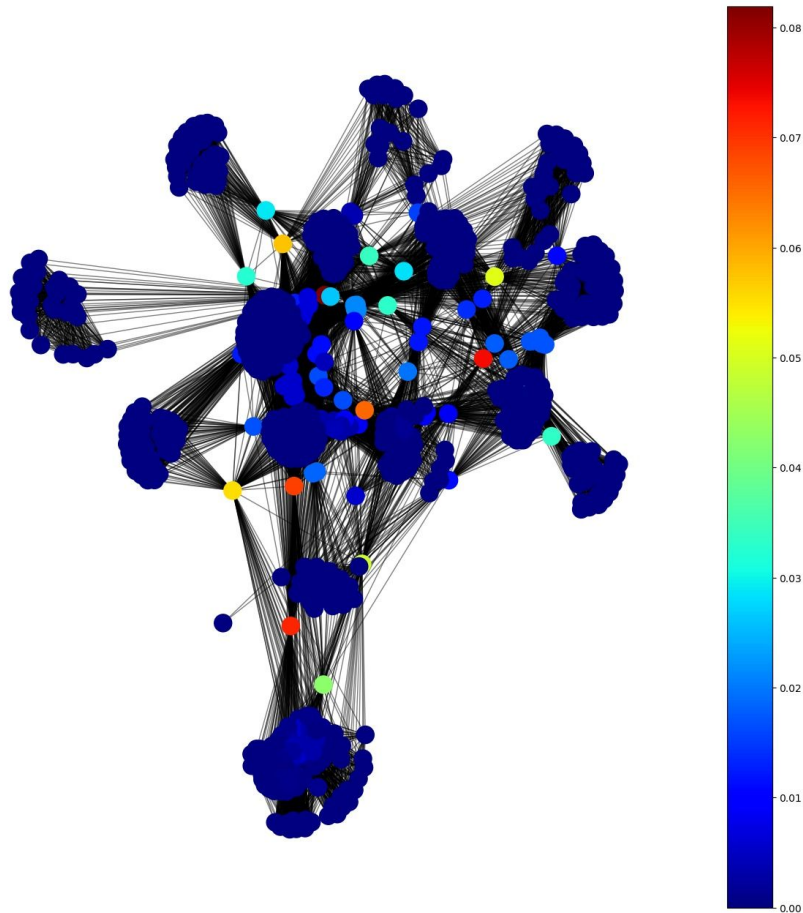
The graph illustrates the degree centrality of nodes in Dataset 1, where **each node represents an entity and edges represent their connections**. The color gradient from blue to red indicates the degree centrality score, with red nodes having the highest number of direct connections and blue the lowest. **Nodes with higher degree centrality are more centrally located and act as key connectors or "hubs" within the network**, while peripheral blue nodes have fewer connections.





Closeness Centrality

The graph visualizes a network where each node's color and size represent its degree centrality—the number of direct connections it has to other nodes. **Nodes with higher degree centrality are shown in warmer colors** (yellow, orange, and red) and are larger in size, indicating they are **highly connected and serve as important hubs** within the network. Conversely, **nodes in cooler colors (blue and green) and smaller sizes have fewer connections**, often appearing on the network's periphery.



Betweenness Centrality

The graph visualizes the betweenness centrality for Dataset 1, where each node represents an entity and edges indicate connections between them. Nodes are colored based on their betweenness centrality values, with the color scale ranging from blue (low centrality) to red (high centrality), as shown by the color bar on the right. Larger nodes appear to have higher connectivity or importance within the network, acting as key bridges or intermediaries between different clusters. The dense web of connections and the presence of several prominent nodes suggest a complex network structure.

Top Artists by Centrality Metrics in Dataset 1

Betweenness Centrality:

- Pink Floyd, Billy Joel, Chris Stapleton, Future, Hozier, Disclosure, Twenty One Pilots, Fleetwood Mac, John Legend, Old Crow Medicine Show.

Degree Centrality:

- Fetty Wap, Kendrick Lamar, Future, Lorde, Frank Ocean, Young Thug, Lil Yachty, 2 Chainz, Kid Cudi, Catfish and the Bottlemen.

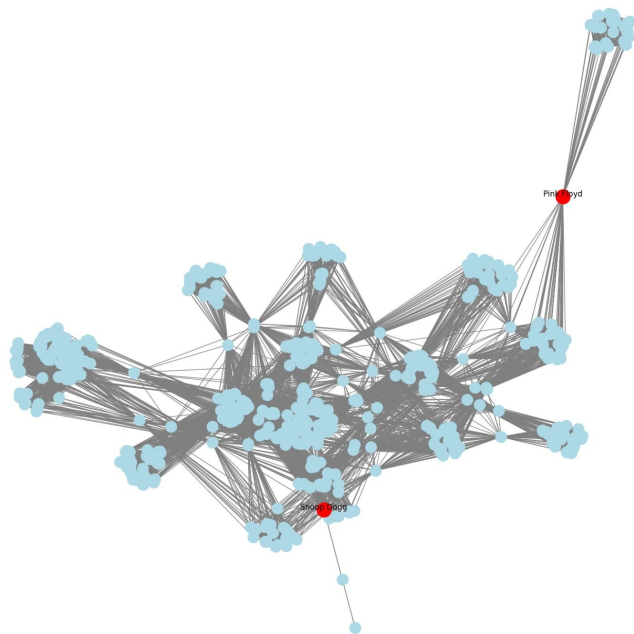
Closeness Centrality:

- The Avett Brothers, Milky Chance, Future, Frank Ocean, Fetty Wap, Kendrick Lamar, Coldplay, Hozier, Young Thug, Lil Yachty.

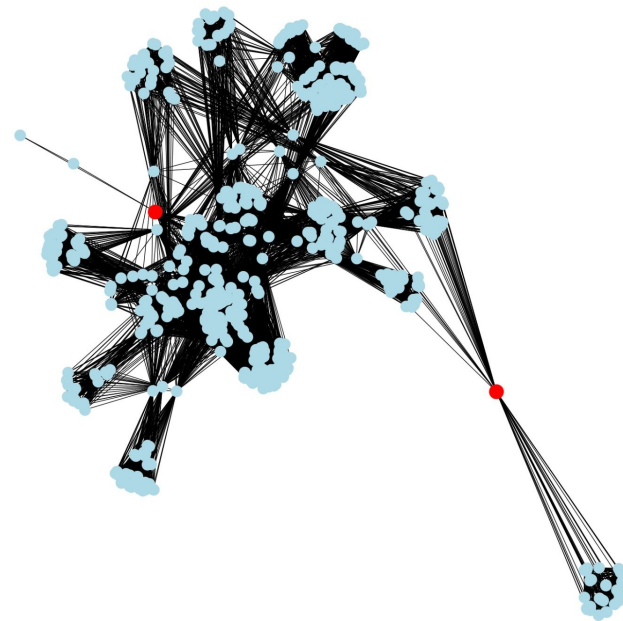


Cut Vertices Graph

Graph with Cut Vertices Highlighted and Labeled



Graph with Cut Vertices Highlighted



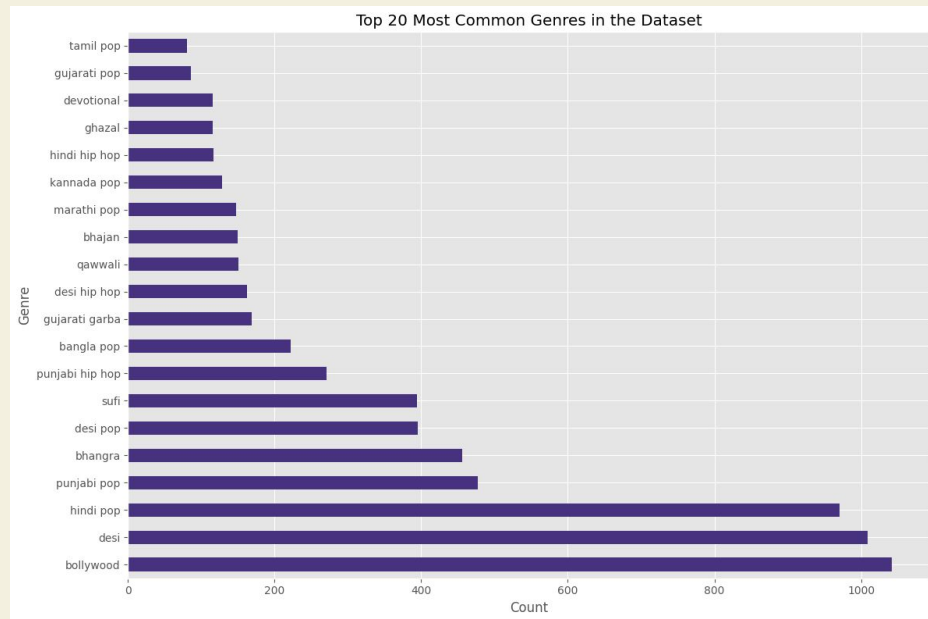
Dataset 2 EDA

The dataset, shaped (1827, 23), is a Spotify music data collection containing 1,827 tracks, each described by 23 features. Key columns include track and album names, artist, release date, duration, popularity, genres, and several musical attributes such as key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and time signature. This rich set of features enables detailed analysis of musical trends, artist styles, and track characteristics, making it valuable for music analytics or recommendation system projects

```
Number of nodes in bipartite graph: 1986  
Number of edges in bipartite graph: 7767  
Number of track nodes: 1827  
Number of genre nodes: 159  
Number of nodes in genre projection: 159  
Number of edges in genre projection: 730
```

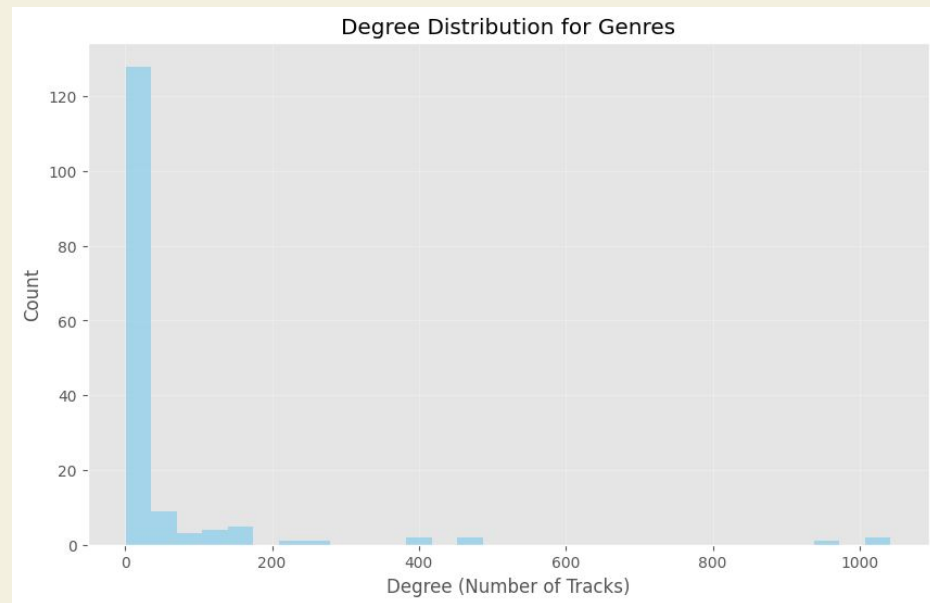
Common Genres

The bar chart and the provided data show that the most common genres in this Spotify dataset are dominated by Indian music categories. "Bollywood" is the most prevalent genre with 1,042 tracks, closely followed by "desi" (1,009) and "hindi pop" (971). Other popular genres include "punjabi pop," "bhangra," "desi pop," and "sufi," each with several hundred tracks. Regional and niche genres like "bangla pop," "gujarati garba," "qawwali," "bhajan," and various language-specific pop styles also have significant representation, reflecting the rich diversity of contemporary Indian music trends in the dataset

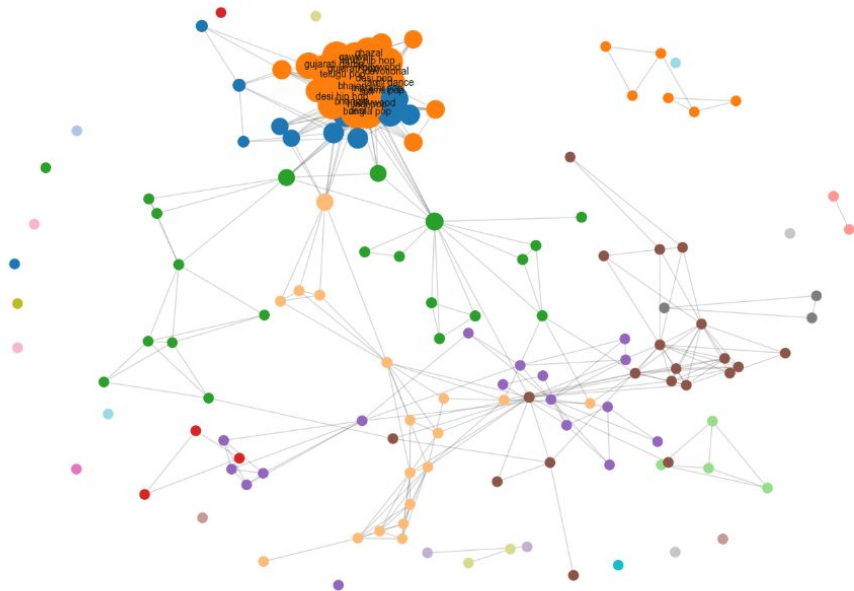


Degree Distribution

This bipartite graph analysis of the Spotify dataset models the relationship between tracks and genres, with 1,827 track nodes and 158 genre nodes connected by 7,561 edges¹. The degree distribution plot for genres shows that while most genres are associated with only a few tracks, a small number of genres—such as "Bollywood" and "desi"—are linked to a large number of tracks, indicating their broad popularity and influence in the dataset. This skewed distribution highlights the dominance of certain genres and the niche presence of others within the Spotify collection.



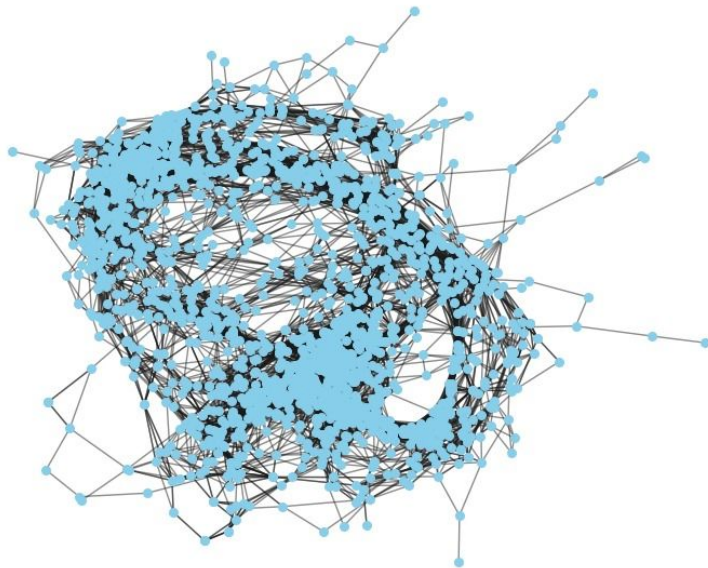
Genre Co-occurrence Network with Communities



Genre Community

This genre co-occurrence network graph visualizes how different music genres are interconnected based on shared tracks, with each color representing a distinct genre community. Central genres like "bollywood," "desi," "pop," and "edm" act as hubs, linking multiple communities, while more isolated clusters and nodes represent niche or less-connected genres. The dense connections among central nodes highlight the strong overlap and blending of popular genres within the Spotify dataset.

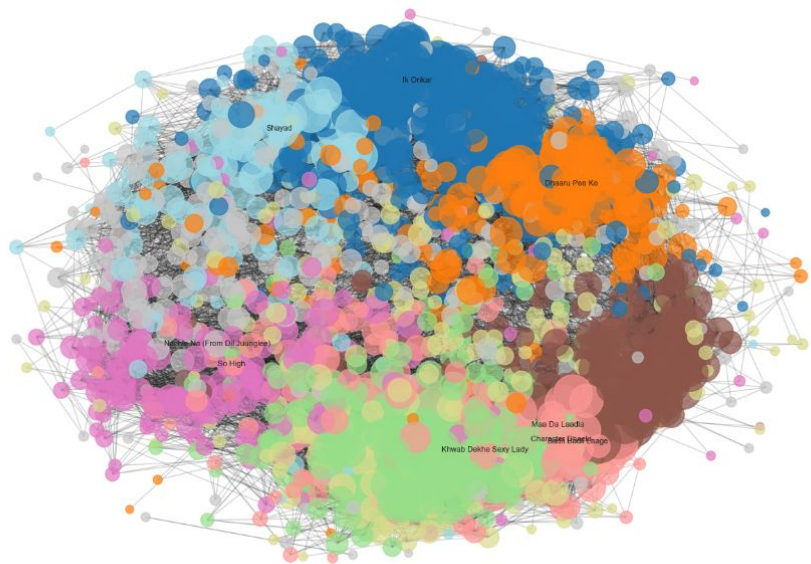
Track Similarity Graph (Largest Component)



Track Similarity

The track similarity graph for this Spotify dataset features **1,827 tracks (nodes) connected by 20,672 similarity links (edges)**, with an average degree of **about 22.6**, indicating that each track is, on average, similar to over 22 others. The network is highly cohesive, as shown by a large main component containing 1,816 tracks and a relatively high clustering coefficient of 0.516, reflecting a strong tendency for similar tracks to form tight-knit groups. There are 10 connected components overall, and the network divides into 19 distinct communities, with **"Jatta Koka" by Kulwinder Billa identified as the most central track**, suggesting it plays a key bridging role within the network.

Mood Network - Songs Grouped by Audio Feature Similarity



Mood Network

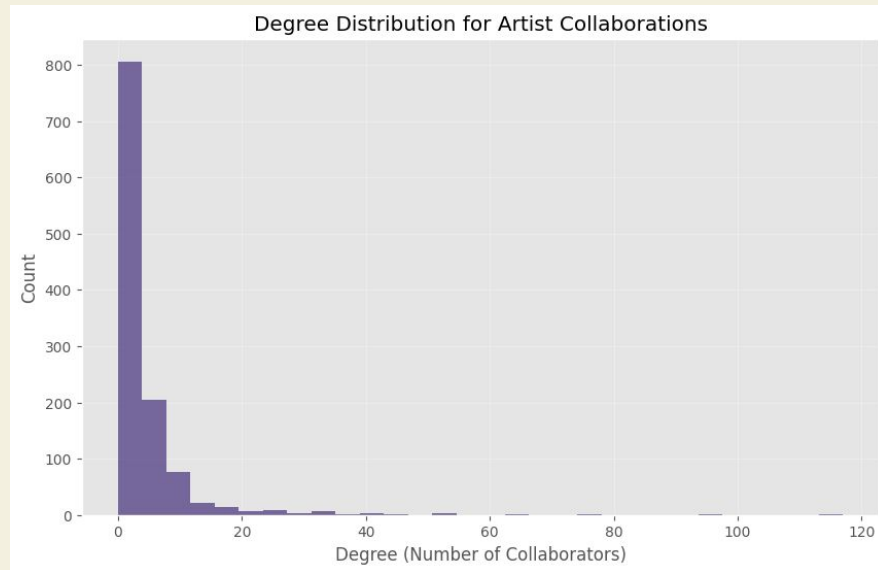
The mood network graph visualizes 1,827 songs grouped by audio feature similarity, forming 9 distinct mood-based communities. With 30,597 edges and a low density of 0.018, the network is well-connected yet sparse, and most songs (1,818) belong to one large cluster. The high modularity score of 0.68 reflects strong, clear separation between mood communities, as shown by the colored clusters in the visualization. If you want a deeper dive into what defines each community, just let me know!

Nodes – songs

Edges – connections based on similarity

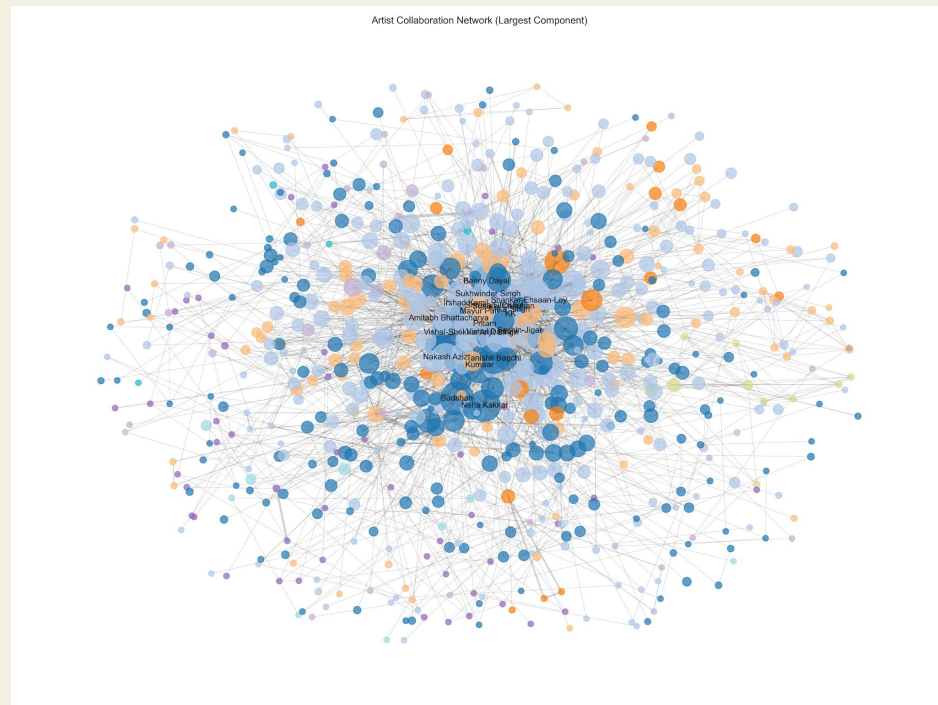
Degree Distribution for Artist Collabs

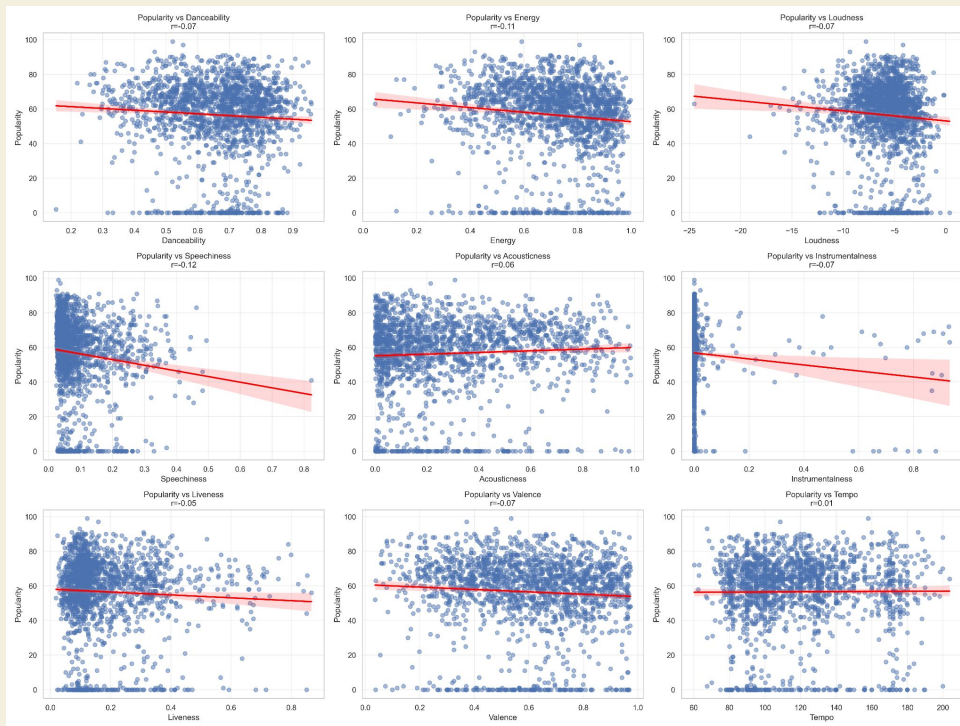
The artist collaboration graph for this dataset includes 1,164 artists connected by 2,501 collaborations. The degree distribution plot shows that most artists have collaborated with only a few others, while a small number of artists have a much higher number of collaborations. This skewed distribution suggests that the music collaboration network is dominated by a few highly connected artists, acting as hubs, while the majority of artists participate in relatively few collaborations.



Artist Collabs Community

The image displays a complex artist collaboration network divided using the Louvain community detection algorithm, **revealing 290 distinct communities with a modularity score of 0.5726**. The network is dominated by five major communities, with the largest (Community 6) containing 146 artists primarily associated with Bollywood, Desi, and Hindi Pop genres. Similarly, Communities 7, 13, and 14 show strong connections to Indian music genres including Bollywood, Desi Pop, and Sufi, while Community 1 stands apart with predominantly Western music genres like pop, Latin, and EDM.



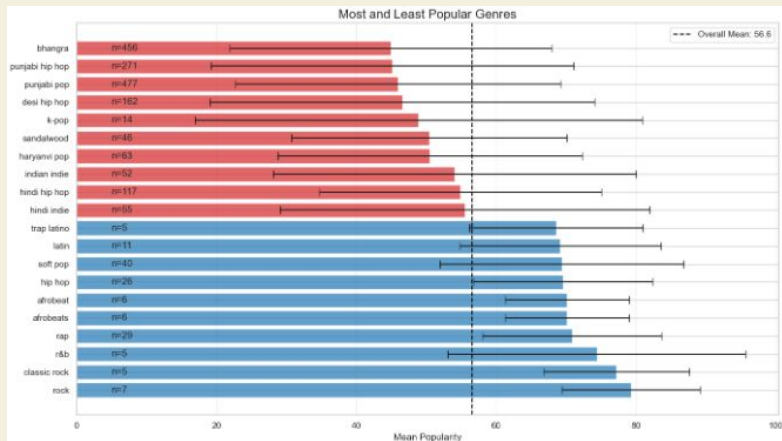
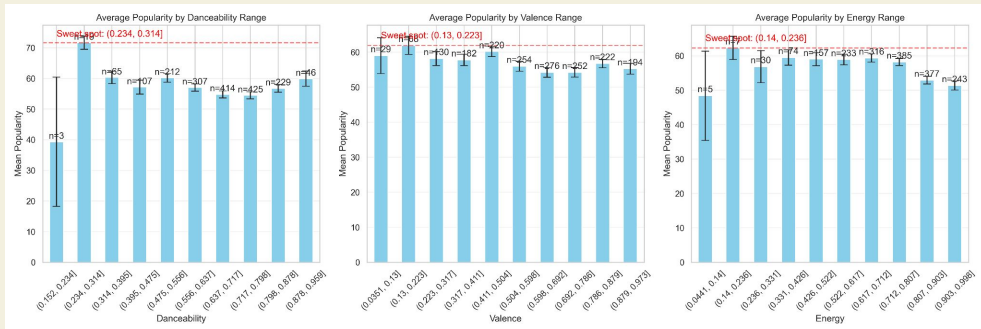


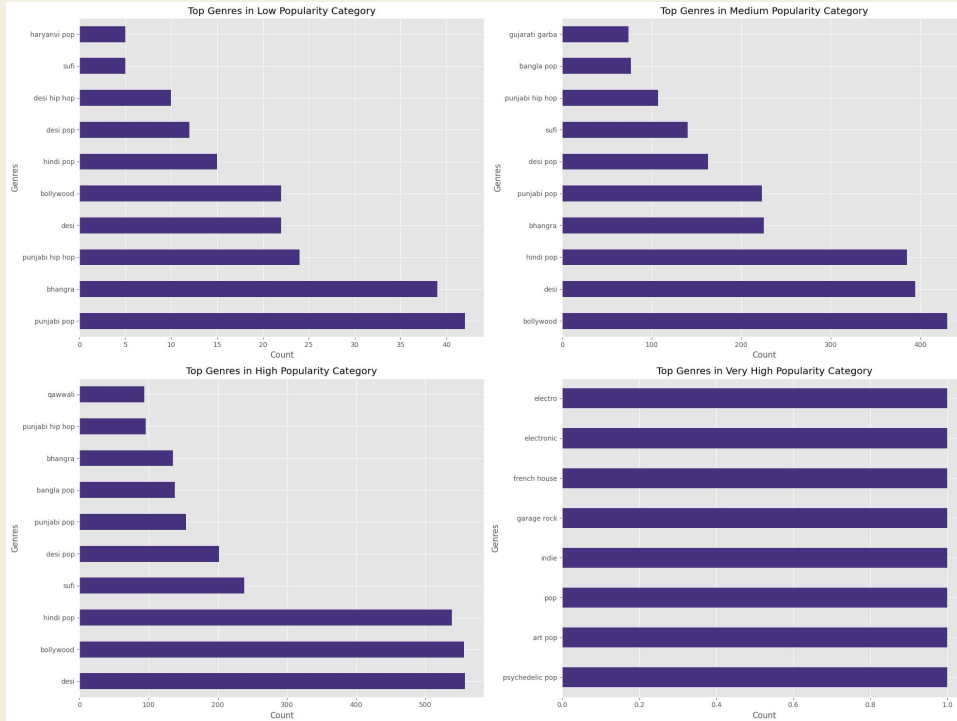
Popularity vs Audio Feature Analysis

The scatter plots illustrate the **relationship between various audio features and track popularity in the Spotify database**. Across features like danceability, energy, valence, tempo, acousticness, speechiness, instrumentalness, liveness, and loudness, the correlations with popularity are generally weak, with most correlation coefficients close to zero or slightly negative. Notably, **features such as energy, speechiness, and instrumentalism show a mild negative correlation with popularity, suggesting that tracks with higher values in these attributes tend to be slightly less popular.**

Popularity vs Audio Feature Analysis

The "Popularity vs Audio Feature Analysis" visualizes how song popularity varies with key audio features—danceability, valence, and energy—using bar charts, and also compares the most and least popular genres. The top charts show that songs with moderate to high danceability, valence (positivity), and energy tend to be more popular, as indicated by higher average popularity scores in those ranges. The bottom chart highlights which genres consistently achieve higher or lower popularity, with clear distinctions between the most and least favored genres.



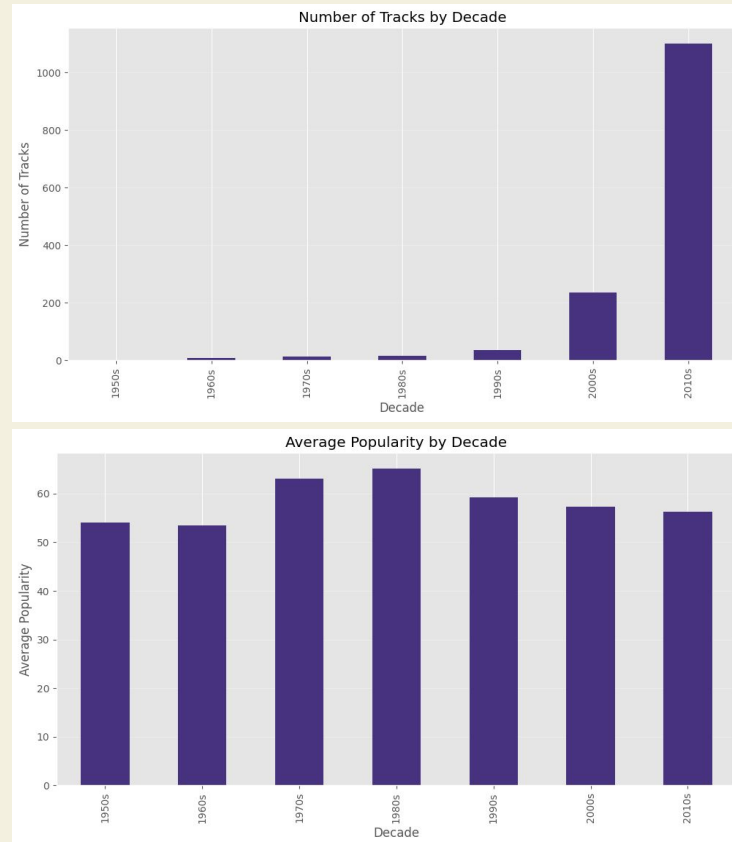


Popularity vs Audio Feature Analysis

The bar charts show the top music genres across four popularity categories in the Spotify database. For low, medium, and high popularity tracks, Indian genres like Bollywood, Desi, Hindi Pop, and Punjabi Pop consistently dominate, with their counts increasing as popularity rises. In contrast, the "very high" popularity category features mostly Western and electronic genres such as pop, indie, electronic, and electro, each with much lower counts, indicating these genres are rare but exceptionally popular when present. This highlights a strong regional preference in the broader dataset, while global pop and electronic genres stand out at the very top of the popularity spectrum.

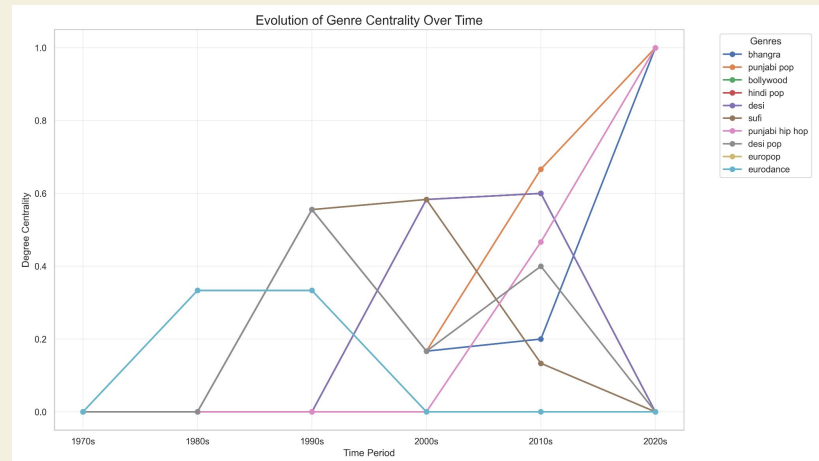
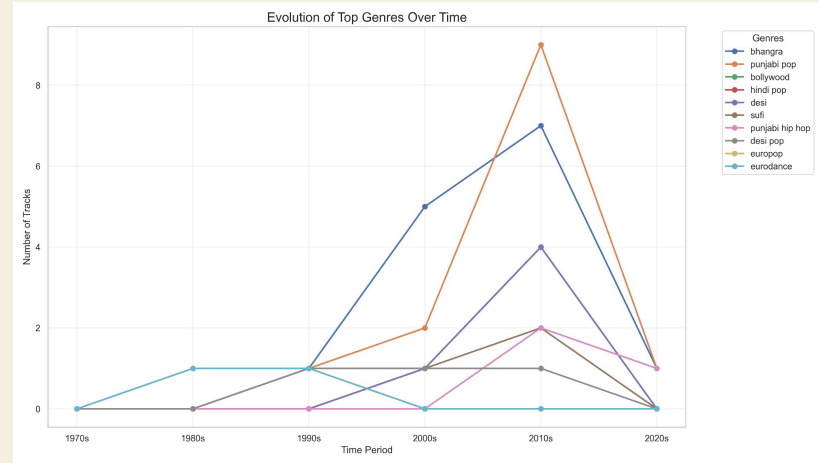
Time-Evolving Network

The charts show that the number of tracks in the Spotify database has grown dramatically over the decades, with a sharp increase in the 2000s and an even larger surge in the 2010s. Despite this growth, average track popularity peaked in the 1970s and 1980s and has declined slightly in more recent decades, suggesting that while more music is being released and added to the platform, newer tracks do not necessarily achieve higher average popularity compared to older decades.



Evolution of top Genres overtime

This illustrate how the prominence and influence of top music genres have shifted over time. The top chart shows that genres like "indie pop" and "pop" experienced significant growth in the 2000s and peaked in the 2010s, while others remained relatively stable or saw only modest increases. The bottom chart tracks genre centrality, revealing that certain genres not only grew in number but also became more central and influential in the music landscape, particularly in recent decades.

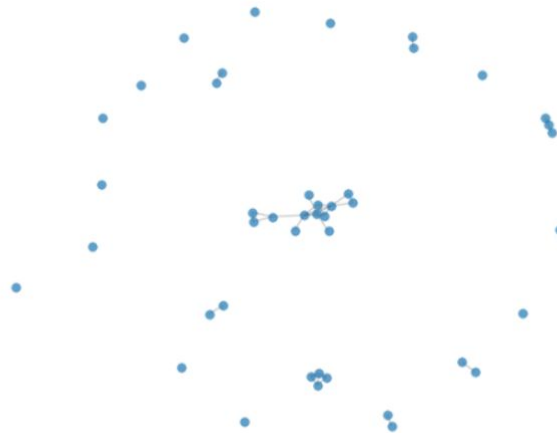


Time-Evolving Network

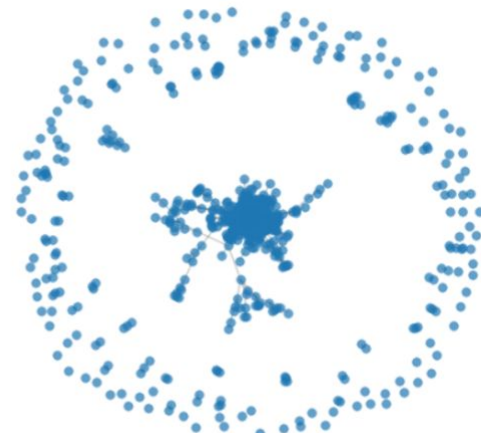
Collab Network: 1970s



Collab Network: 1990s



Collab Network: 2010s



1. Introduction
2. Datasets Used and Work Done
3. Drawing a Conclusion

CONCLUSION

- Mapped dynamic relationships between artists and genres using network science, revealing strong community structures and interconnected genre hubs.
- Indian genres like Bollywood, Desi, and Hindi Pop dominate the dataset, especially in collaborative and popular track communities.
- Audio features such as danceability, energy, and valence show weak correlations with track popularity, indicating that musical success is influenced by multiple factors.
- Track volume on Spotify has surged in recent decades, but average popularity has not increased, suggesting that more music does not guarantee greater influence.
- Network analysis uncovers hidden patterns in music trends, artist collaborations, and genre evolution, offering valuable insights for music analytics and recommendation systems